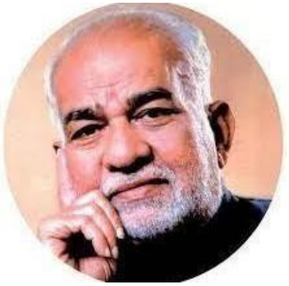
BVRIT HYDERABAD College of Engineering for Women

(UGC Autonomous)

R&D SHOWCASE 2026



RythuMitra: A Multilingual Voice-First AI Farming Companion

ABSTRACT

RythuMitra is a voice-first AI farming system that enables multilingual interaction and real-time crop disease detection using deep learning. It provides personalized recommendations and integrates blockchain for secure scheme applications. With access to real-time market and weather updates, it supports efficient farming and promotes smart agriculture.

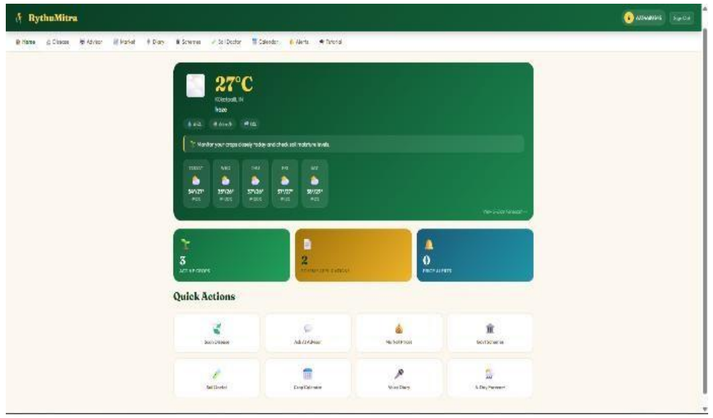
UNIQUENESS

• Voice-first interaction	• Multilingual support
• Scheme support	• Farmer-friendly UI

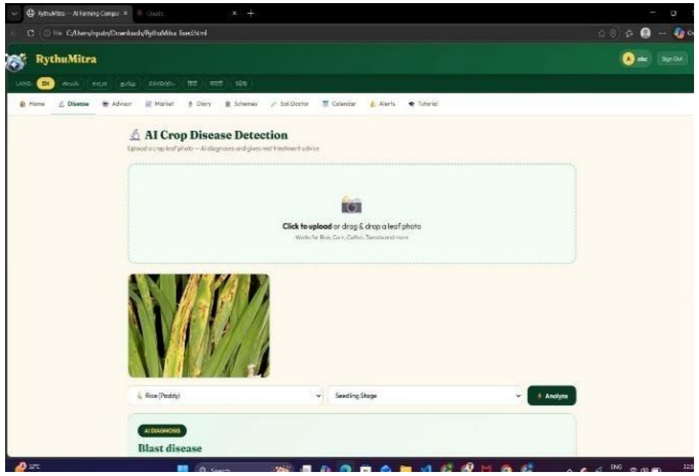
METHODOLOGY

The methodology uses CNN models for crop disease detection, optimizing predictions through image preprocessing and classification. NLP-based voice interaction enables multilingual communication, while blockchain ensures secure scheme processing. This helps in accurate and efficient farming decisions.

Dashboard



Crop Disease Detection



SOCIETAL USE

RythuMitra supports sustainable agriculture by oenabling early crop disease detection, reducing losses, and improving farmer productivity. Its voice-first multilingual design makes technology accessible to low-literacy users, while blockchain ensures transparent scheme access. The system contributes to digital empowerment of farmers and supports Sustainable Development Goals like Zero Hunger and Industry Innovation

CONCLUSION

RythuMitra presents an innovative solution for modern agriculture by combining artificial intelligence, blockchain, and voice-based technologies. The system enhances accessibility, improves decision-making, and provides real-time insights to increase productivity and support smart, sustainable farming practices.

REFERENCES

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